TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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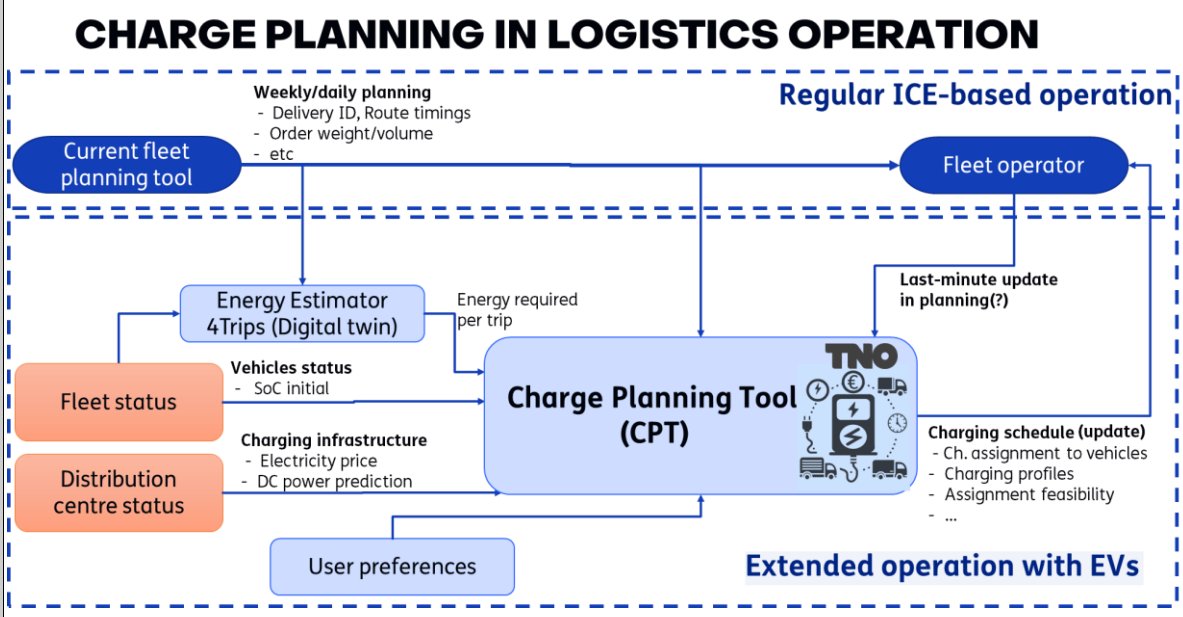
2. Technology Overview

Title of Proposed Technology	Digital twin optimised charge planning software for commercial EV fleets and charging hubs			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-generation Energy (√), Bio-health ( ), Aerospace ( ), Quantum ( ), Advanced Robotics & Manufacturing ( ), AI (√), Cybersecurity (), Hydrogen ( ), Semiconductor materials ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communications		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Internet SW	Heavy Electrical Equipment	Power conversion equipment
Concept of Proposed Technology	Core concept: An integrated fleet charge planning engine that uses digital twins of vehicles and batteries to generate feasible, cost-optimal charging schedules under real-world constraints. Key elements: 1) Charge Planning Tool (CPT) algorithms to schedule charging for non-homogeneous EV fleets across multiple charging opportunities while considering charger-vehicle compatibility, limited grid capacity and charging operation cost components. [R1]			

	2) Battery Digital Twin models that predict per-vehicle charging power profiles and charge time while accounting for battery aging and uncertainty. [R2] 3) Fleet Planning Tool to Digital Twin interface for periodic vehicleStatus updates and vehiclePrediction outputs (e.g., arrivalSoC, predictedPowerProfile, timeToFull), enabling closed-loop planning and re-planning. [R4] 4) Standardised integration with charging management platforms using OCPP and energy management coordination using OpenADR concepts as referenced in CEH interface work. [R4] 5) Validation and calibration using TNO Battery Lab and powertrain testing facilities (climate-controlled testing up to battery pack and vehicle level; 300 kW bidirectional cycling). [R5, R6, R7] Potential uses: depot and hub charging optimisation for logistics operators, municipal fleets and service providers; congestion-aware charging; design and what-if analysis for new charging hubs. 
Rationale for Joint R&D Project	Need: Commercial EV fleets face operational risk and higher cost when charging is planned separately from routing, especially under constrained grid connections and limited chargers. TNO has demonstrated a Charge Planning Tool that incorporates charging into logistics planning to support investment and operational decisions. [R3] Collaboration rationale: The Korean SME partner has an operational full-stack EV charging software stack (CSMS, OCPP API, CRM integration and fleet applications) and seeks to add or improve digital twin based optimisation capability. TNO contributes validated optimisation algorithms, digital twin modelling methods and test and validation facilities. Joint R and D

	enables rapid integration into deployable software, plus validation on both Korean and European use cases and data. Why TIPA: International joint development de-risks market entry and standardisation, and accelerates transfer from research prototypes to a commercial-grade toolchain.
Key Words	1. Digital twin 2. Charge planning 3. Fleet optimization 4. Smart charging 5. Battery aging

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	Objective 1: Integrate TNO's Charge Planning Tool (CPT) concepts with the Korean SME partner's charging management and fleet applications, enabling practical, automated day-ahead and intra-day charging plans for mixed fleets under grid and charger constraints. [R1, R3] Objective 2: Implement and validate battery and vehicle digital twins to predict charge power profiles, charge time and state-of-charge evolution during operations and charging, improving planning robustness under aging and uncertainty. [R2, R8] Objective 3: Demonstrate closed-loop optimisation with standardised data interfaces (vehicleStatus and vehiclePrediction) between Fleet Planning Tool and Digital Twin, aligned with work in the Charging Energy Hubs programme. [R4] Objective 4: Validate the integrated solution using laboratory testing and pilot data, and deliver a deployable API and reference implementation suitable for commercialisation and further scaling.			
R&D Target Specifications	Target KPI 1 (prediction): Battery Digital Twin charge-time prediction error below 2% on representative charging sessions, consistent with published results on TNO battery digital twin methods. [R2] Target KPI 2 (planning): Generate feasible charging schedules for non-homogeneous fleets with less chargers than vehicles and with grid capacity constraints, as required in CPT research. [R1] Target KPI 3 (cost and congestion): Reduce peak depot power demand by at least 10% versus uncontrolled charging, and reduce energy cost by at least 5% versus a baseline rule-based schedule, using tariff and grid constraints (project targets to be validated during pilots). Target KPI 4 (runtime): Produce day-ahead schedules within 5 minutes for fleets up to 200 vehicles, enabling operational use (project target; baseline performance reported for related real-time scheduling algorithms). [R9] Target KPI 5 (integration): Support OCPP based charger control and ISO 8601 time formatting, and publish an API specification for integration into CSMS and EMS components. [R4]			
R&D Plan	Assumptions (for planning): 36 months total duration, structured in 3 phases (P1: months 1-12, P2: months 13-24, P3: months 25-36). These will be aligned with the final TIPA workplan. **WP1 Requirements, data and architecture (M1-M6):** • Gather operational requirements and datasets (routes, charging sessions, grid connection,			

charger inventory).

- Define system architecture and interfaces between SME platform, Fleet Planning Tool and Digital Twins, based on the CEH interface definition (vehicleStatus, vehiclePrediction, planningFeasibility, predictedPowerProfile). [R4]
- Define cybersecurity, data governance and anonymisation approach for pilot data exchange.

WP2 Battery Digital Twin and charging profile prediction (M1-M18):

- Implement battery digital twin method for charge profile generation (adaptive, low training cost) and calibrate to fleet data, following Rath and Wilkins (VPPC 2024). [R2]
- Extend to include state-of-health and aging indicators for planning cost terms, and quantify uncertainty bounds.
- Validate DT predictions using existing datasets and laboratory tests.

WP3 Charge planning optimisation engine (M4-M24):

- Implement charge planning optimisation for mixed fleets using CPT principles (charger compatibility, multiple charge opportunities, grid constraints and cost components). [R1]
- Integrate DT predictions into planning loop, aligning with fleet DT integration approaches demonstrated at EVS38 2025. [R8]
- Add congestion-aware control hooks for charger setpoints and session authorisation.

WP4 Laboratory validation and model-in-the-loop testing (M6-M30):

- Validate charging profiles and thermal constraints using TNO Battery Lab and powertrain testing facilities (climate-controlled testing and 300 kW cycling). [R5, R6, R7]
- Stress-test optimisation under realistic disturbances (delays, route changes, temperature) and evaluate robustness.

WP5 Pilot deployment and demonstration (M18-M36):

- Deploy to a pilot fleet or charging hub environment via SME platform integration (CSMS and fleet app).
- Measure KPI performance (cost, peak demand, schedule feasibility, prediction accuracy).
- Deliver final software package: API specification, reference implementation, and deployment documentation.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (Research Institute) responsibilities: • Provide scientific leadership on charge planning optimisation and digital twin modelling. • Provide reference implementations and know-how from CPT research and CEH tool interfaces. [R1, R4] • Develop and validate battery digital twin charge profile prediction methods. [R2] • Execute laboratory validation at TNO facilities (Battery Lab and powertrain test centre), including climate-controlled testing and high-power cycling. [R5, R6, R7] • Support dissemination: publications, workshops, and contribution to interoperable interface specifications.
Expected R&R of Korean SME Partner	Korean SME partner responsibilities: • Provide the commercial software stack (CSMS, OCPP integration, user interfaces, CRM integration and fleet applications) and operational deployment capability (as shared in partner-provided non-public materials, Dec 2025). • Provide anonymised operational data (charging sessions, fleet schedules) and access to pilot environments in Korea. • Implement product-grade integration of optimisation and DT services into the SME platform, including CI/CD, monitoring and customer support functions. • Lead business-case development, customer engagement and commercialisation planning for the Korean and international market.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected technical outcomes: • Integrated charge planning and digital twin software module with documented APIs for CSMS and fleet software. • Validated battery digital twin models for charge profile and time-to-full prediction, with quantified uncertainty. [R2] • Demonstrated fleet-level charging optimisation under grid constraints with measurable reduction in peak demand and cost (pilot results). [R1, R3, R8] Expected impact: • Improved depot and hub utilisation, enabling more EVs on existing grid connections and fewer costly grid upgrades. • Enhanced operational reliability and lower total cost of ownership for fleet operators via better planning and battery-aware charging. • Strengthened Korea Netherlands collaboration and pathway to standardised, interoperable smart charging solutions.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization